Arista Access Node MAR-ACC-30 QoS / Congestion Event Resolution SOP

Node: MAR-ACC-30 | **Vendor:** Arista | **Severity:** Minor | **Alarm Code:** ALARM_6887

Description

This Standard Operating Procedure (SOP) is designed to address QoS / Congestion Event failures on the Arista access node MAR-ACC-30, located at Marathahalli Core. The failure domain is triggered by microbursts, interface tail drops, and QoS queue starvation, resulting in the ALARM_6887 Minor severity alarm. This SOP is specific to the Arista vendor equipment and the Access role. The underlying protocol and hardware mechanics responsible for this failure domain can be attributed to the Arista EOS software's prioritization of packets based on DSCP (Differentiated Services Code Point) values, which may lead to queue starvation and congestion when the QoS policies are misconfigured or not properly optimized. Furthermore, the Arista 7120S series switch's hardware capabilities, such as the 400G QSFP-DD ports, may also contribute to congestion when handling high-bandwidth traffic. This SOP outlines the necessary diagnostic and resolution steps to identify and rectify the root cause of the QoS / Congestion Event failure, ensuring minimal disruption to network services.

Pre-requisite Checks

- Verify that all network personnel have been notified and informed about the ongoing maintenance.
- Dispatch remote hands to Marathahalli Core to facilitate physical access to the affected node.
- Check the status of redundant links to ensure that they are up and operational.
- Confirm that the network is in a stable state and no other severe alarms are present.

Potential Root Causes

Cause	Probability
Misconfigured or inefficient QoS policies	High
Insufficient buffer space on the interface	Medium
Hardware limitations on the 7120S series switch	Low

Diagnostic Commands & Expected Outputs

```
Command: show qos statistics interface 1/1 detail, Expected Output: Interface: 1/1, QoS Statistics:, Description: This command displays detailed QoS statistics for the interface 1/1, including the number of packets and bytes transmitted, as well as the number of drops and errors. This information can help identify potential congestion and queue starvation issues.

Command: show qos policy-map interface 1/1, Expected Output: Policy Map: policy_map_1_1, Interface: 1/1, Description: This command
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```
displays the QoS policy map applied to interface 1/1, including the class maps, match criteria, and actions. This information can help identify misconfigured or inefficient QoS policies.

Command: show interface 1/1 counters, Expected Output: Interface: 1/1, Counters:, Description: This command displays the interface counters for interface 1/1, including the number of packets and bytes transmitted, as well as the number of drops and errors. This information can help identify potential hardware limitations on the 7120S series switch.
```

Resolution Steps

Step 1: Step: Verify QoS policy configuration, Command: show qos policy-map interface 1/1, Description: Verify that the QoS policy map is correctly configured and applied to interface 1/1. Check for any misconfigured or inefficient policies that may be contributing to the congestion and queue starvation issues.

Step 2: Step: Adjust QoS policy, Command: configure terminal policy-map policy_map_1_1 class class-default set dscp 46 exit exit, Description: Adjust the QoS policy map to prioritize packets with DSCP value 46 (EF) above other traffic. This can help alleviate congestion and queue starvation issues.

Step 3: Step: Increase buffer space, Command: configure terminal interface 1/1 buffer 1000 exit, Description: Increase the buffer space on interface 1/1 to 1000 packets. This can help alleviate congestion and queue starvation issues by providing more buffer space for incoming packets.

Post-Check Verification

```
Command: show qos statistics interface 1/1 detail, Expected Output: Interface: 1/1, QoS Statistics:, Description: Verify that the QoS statistics have improved and the number of drops and errors has decreased.

Command: show interface 1/1 counters, Expected Output: Interface: 1/1, Counters:, Description: Verify that the interface counters have improved and the number of drops and errors has decreased.
```

Roll-back Procedures

- Step: Revert QoS policy changes, Command: configure terminal undo policy-map policy_map_1_1, Description: Revert the QoS policy changes made in step 2 to ensure that the original policy is restored.

- Step: Revert buffer space changes, Command: configure terminal undo interface 1/1 buffer 1000, Description: Revert the buffer space changes made in step 3 to ensure that the original buffer space is restored.